

Establishing a 5G Testbed Using Open Source Technology

Open source plays a very crucial part in the world of technology. Every piece of technology today is either fundamentally dependent on it to leverage its growth or to become completely operational. Let us see how it fares with the development of the 5G testbed.

The open source effort to establish a 5G testbed is part of a project that was founded by the Department of Telecommunications of the government of India. It has been running for about four years now. There were nine principal investigators on this project.

The project itself officially ended on 1st January 2022. Now, at the Indian Institute of Science in Bengaluru we are on to the maintenance of the project where we maintain the testbed and encourage its usage and do minor developments.

The following is the overview of different aspects of what this project is all about. There are fundamentally four parts to it. Let us dive in and understand about these attributes.

1. System development in sub 6GHz (FR1)

The 6-gigahertz (GHz) development

platform is the portion that is actually open source. We have primarily looked at physical layer enhancements and implemented the split six architecture, also called algorithms, and integration with different radio units. We have developed antenna arrays 16 by 4 that are 6499 arrays with an up-down converter and transceiver for the sake of completeness, while testing it alongside extensively in an antiquated chamber.

2. Vehicle-to-X (V2X) platform

This is a Long Term Evolution (LTE) based platform. It is a fully functional platform, which can perform end-to-end latency measurements. We have enhanced the MAC schedulers in LTE and reduced the latency of a native implementation of LTE by a factor between 5 and 10. Also, the remote driving capabilities have been demonstrated on campus.

3. Visible light communications (VLC) system development

This is a complementary technology to 5G and not a mainstream one. We have built a communication system which is able to deliver about 4.8 gigabytes per second using OFDM. It also incorporates beam steering and has technical knowledge to create its backbone.

4. Features of Sub-6 system

We are using a generic radio platform based on the universal serial radio port. This can be operated anywhere up to 6GHz but is primarily focused on the 3.3GHz to 3.8GHz frequency band. The other features are all supported by our testbed, including 30kHz carrier spacing. It has around 100MHz bandwidth and up to 256 QAM. The value proposition of this testbed is that it is a fully open sourced platform, so every part of the 5G radio stack is available to be modified. It has been well tested with USRP X310 and B210 radios as well as with commercial phones like the OnePlus 9 Pro 5G and Oppo A74.

Lab system: connection model

Essentially, we have USRP radios that are connected to a personal computer (PC) where the gNB1 software stack will be running in a monolithic implementation. Then they are

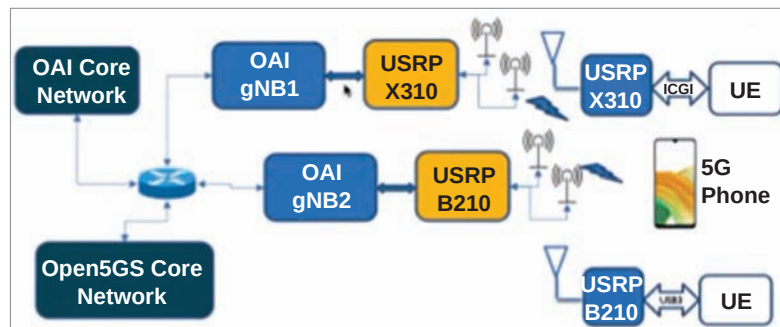


Figure 1: Lab system: connection model